AYE
TECH HUB

• HVAC ENGINEERING • LESSON 01

Introduction to HVAC Engineering

What HVAC Is, Why It Matters & How It Works

Lesson 1 of 30 | HVAC Engineering Program | Beginner

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- HVAC defined — Heating, Ventilation & Air Conditioning
- The 4 core functions and what each system controls
- Key components — compressors, AHUs, ducts, chillers
- The refrigeration cycle — how cooling actually works
- HVAC career pathways in the engineering industry

01 | What Is HVAC Engineering?

HVAC stands for **Heating, Ventilation, and Air Conditioning**. It is the engineering discipline responsible for controlling the indoor climate, air quality, and thermal comfort of buildings — from small apartments to hospitals, data centres, and industrial plants. Every building you enter has an HVAC system working behind the scenes.

HVAC engineers design, size, install, and maintain these systems. They work alongside structural, electrical, and plumbing engineers to ensure buildings are safe, efficient, and comfortable year-round. In the MEP (Mechanical, Electrical & Plumbing) industry, HVAC is the "M" — the mechanical backbone of every building services project.

System	What It Controls	Key Equipment	Where It Matters
Heating	Indoor temperature in cold conditions	Boilers, heat pumps, radiators, FCUs	Offices, hospitals, cold-climate buildings
Ventilation	Fresh air supply and exhaust of stale air	AHUs, fans, dampers, ERVs	All occupied spaces — required by code
Air Conditioning	Cooling and humidity control	Chillers, split ACs, VRF systems, FCUs	Offices, malls, data centres, homes
Air Quality	Filtration, CO2 control, pressure management	HEPA filters, CO2 sensors, UV systems	Hospitals, labs, clean rooms, schools

INFO — HVAC Is the Largest Energy Consumer in Buildings

HVAC systems account for 40–60% of total building energy consumption. A well-designed HVAC system can reduce energy bills by 30–50% compared to a poorly designed one. This is why HVAC knowledge is critical for engineers focused on green building and energy efficiency.

02 | The Four Core Functions — What Each System Does

Every HVAC system performs one or more of these four functions. Understanding each one separately is the foundation of HVAC engineering — before you can design a system, you must know exactly what it is trying to achieve.

Function	Engineering Goal	Design Input	Failure Consequence
Heating	Raise indoor air temperature to comfort level (20–22°C typical)	Heat loss calculation, heating load (kW)	Occupant discomfort, frozen pipes, condensation
Ventilation	Supply fresh outdoor air, remove CO2 and odours	Occupancy rate, air changes per hour (ACH)	Poor air quality, sick building syndrome
Air Conditioning	Lower temperature and control relative humidity (40–60% RH)	Cooling load (kW), latent + sensible heat	Heat stress, equipment overheating, mould growth
Air Quality Control	Filter particulates, control pressure and pathogens	Filter grade (MERV/HEPA), room pressure (Pa)	Infection spread, equipment contamination

03 | Key HVAC Components — What You Will Work With

HVAC systems are made up of interconnected components. Every engineer entering the field must be able to identify these components, understand their function, and know where they appear in residential, commercial, and industrial projects.

Component	Function	Found In
Compressor	Pressurises refrigerant to drive the cooling cycle	Split ACs, VRF units, chillers
Condenser	Releases heat from refrigerant to outside air or water	Outdoor units, cooling towers
Expansion Valve (EEV)	Reduces refrigerant pressure, triggers evaporation and cooling	All refrigeration-based systems
Evaporator / FCU	Absorbs heat from indoor air, producing the cooling effect	Indoor units, fan coil units, AHUs
Air Handling Unit (AHU)	Conditions and distributes air via fans, coils, and filters	Large commercial and industrial buildings
Duct System	Distributes conditioned air throughout the building	All centrally ducted HVAC systems
Chiller	Chills water for distribution to AHUs and FCUs across a building	Hotels, malls, hospitals, towers
Cooling Tower	Rejects heat from chilled water loop to the atmosphere	Central chiller plant systems
Thermostat / BMS	Controls and monitors system operation for comfort and efficiency	All modern HVAC installations

04 | The Refrigeration Cycle — How Cooling Actually Works

Every air conditioning system — from a wall-mounted split AC to a 1,000-ton chiller plant — operates on the same principle: the **vapour compression refrigeration cycle**. Understanding this cycle is fundamental to HVAC engineering. It has four stages.

Stage	Process	Refrigerant State	Component	What Happens
1. Compression	Low-pressure gas is compressed	Low-pressure gas → High-pressure hot gas	Compressor	Temperature and pressure rise sharply. Energy input here drives the whole cycle.
2. Condensation	Hot gas loses heat to outside air	High-pressure hot gas → High-pressure liquid	Condenser	Heat is rejected outside. This is why outdoor units blow hot air.
3. Expansion	Liquid passes through a valve, pressure drops	High-pressure liquid → Low-pressure cold liquid/gas mix	Expansion Valve	Temperature drops rapidly — refrigerant becomes very cold.
4. Evaporation	Cold refrigerant absorbs heat from indoor air	Low-pressure cold mix → Low-pressure gas	Evaporator / FCU	Indoor air loses heat to refrigerant. This is the actual cooling effect.

KEY CONCEPT — Heat Is Moved, Not Created or Destroyed

Air conditioning does not create cold air. It moves heat from inside to outside using refrigerant as the transport medium. The refrigeration cycle is a heat pump operating in cooling mode. The same cycle reversed is how heat pumps provide heating in winter.

05 | HVAC System Types — From Homes to High-Rise Buildings

System Type	Scale	How It Works	Typical Application
Split AC	Room / small apartment	One outdoor unit + one indoor unit. Direct refrigerant to FCU.	Homes, small offices, server rooms
Multi-split / VRF	Medium to large buildings	One outdoor unit serves multiple indoor FCUs. Variable refrigerant flow.	Offices, hotels, retail, residential towers
Centralised Chilled Water	Large commercial and industrial	Chiller produces cold water → distributed to AHUs and FCUs via pipes.	Hospitals, malls, universities, airports
Package Unit	Rooftop / commercial	All components (compressor, coil, fan) in one unit. Ducted to floors below.	Single-floor offices, warehouses, retail
Chiller + Cooling Tower	Very large / district cooling	Water-cooled chiller rejects heat via cooling tower. Highly efficient at scale.	High-rise towers, data centres, industrial plants

06 | HVAC Career Pathways

Role	Core Responsibilities	Key Skills	Avg Salary (USD/yr)
HVAC Design Engineer	Load calculations, equipment selection, drawings	HAP/IES software, AutoCAD, ASHRAE standards	\$55,000 – \$90,000
HVAC Site Engineer	Supervise installation, commissioning, testing	Site reading, coordination, punch lists	\$45,000 – \$75,000
MEP BIM Coordinator	Model HVAC systems in Revit, clash detection	Revit MEP, Navisworks, IFC coordination	\$60,000 – \$95,000
Energy Auditor	Assess building energy performance and HVAC efficiency	Energy modelling, ASHRAE 90.1, BMS data	\$55,000 – \$85,000
HVAC Project Manager	Manage HVAC subcontracts, schedule, budget	Project management, cost control, specs	\$70,000 – \$120,000

Reflection:

"Think of any building you visit regularly — an office, a hospital, a mall, or your home. What HVAC systems do you think are operating there right now? What happens to the people inside if those systems fail for 24 hours?"

Action checklist:

Next up — HVAC-002:
The Refrigeration Cycle — a deep dive into how compressors, condensers, expansion valves, and evaporators work together.

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